PROPERTIES OF DURATION

Learning Outcome

- explain how a bond's maturity, coupon, and yield level affect its interest rate risk

The Macaulay and modified yield duration statistics for a traditional fixed-rate bond are primarily functions of the bond's features: its time-to-maturity, its coupon rate, its yield-to-maturity, and the fraction of the current coupon period that has elapsed. The closed-form formula for Macaulay duration, presented in a prior lesson and again here, is useful in demonstrating the characteristics of duration (the same characteristics hold for modified duration, money duration, and price value of basis point).

$$MacDur = \left\{ \frac{1+r}{r} - \frac{1+r+[N \times (c-r)]}{c \times \left[(1+r)^N - 1\right] + r} \right\} - \frac{t}{T}$$

The relationship between a bond's duration and its features (r, c, N, and t/T) are summarized in Exhibit 4.

Exhibit 4:

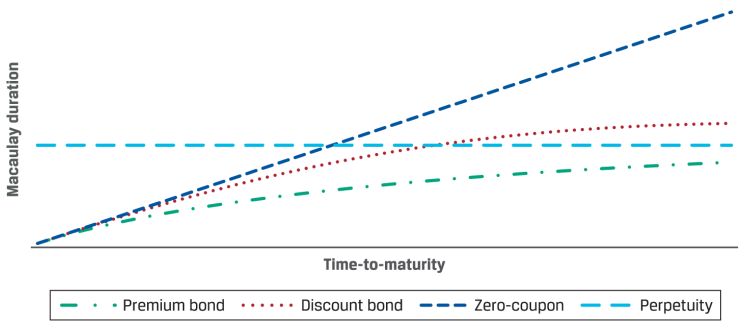
Properties of Yield Duration Statistics

	Effect on duration (interest rate risk) from an *increase* in feature
Coupon rate, c	↓ (Inverse relationship)
Yield to maturity, r	↓ (Inverse relationship)
Time-to-maturity, T	↑ (Direct relationship)
Fraction of current coupon period that has elapsed, t/T	↓ (Inverse relationship)

Exhibit 5 illustrates the relationships between Macaulay duration and time-to-maturity for premium, discount, zero-coupon, and perpetual bonds (the same characteristics hold for modified duration, money duration, and price value of basis point).

Exhibit 5:

Properties of Macaulay Duration



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The same pattern holds for the yield-to-maturity, so,

- *a lower yield-to-maturity increases the weighted average of the time to receipt of cash flow and thus increases duration*.

The reason is that lower coupons and lower yields increase the weight of the maturity value or final cash flow and reduce the weight of nearer-term cash flows in the bond price.

- *Longer times-to-maturity typically correspond to higher duration*.

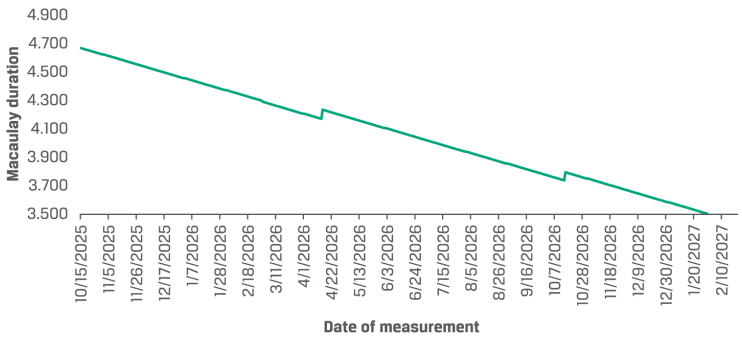
This pattern always holds for bonds trading at par or at a premium. In the closed-form formula, the second expression within brackets is positive for premium and par bonds, because the coupon rate (c) is greater than or equal to the yield-to-maturity (r), so $(c - r) \geq 0$, whereas the denominator is always positive. Therefore, the Macaulay duration is always less than $(1 + r)/r$, and it approaches that threshold from below as the time-to-maturity increases.

A curious result is illustrated in Exhibit 5 for *discount bonds*: at some point, when the time-to-maturity is high enough, the Macaulay duration will exceed $(1 + r)/r$, reach a maximum, and then decrease. This pattern develops when the number of periods (N) is large and the coupon rate is below the yield-to-maturity, so $(c - r) < 0$. Then, the numerator of the second expression within braces can become negative. The implication is that for long-dated bonds trading at a discount to par, interest rate risk could be lower than that of a shorter-term bond.

For a constant yield-to-maturity (r), the expression in braces in the closed-form formula is unchanged as time passes *during the period*. Therefore, the Macaulay duration decreases smoothly as time passes and then jumps upward after the coupon is paid, which creates a "saw-tooth" pattern. This pattern for the BRWA bond with a constant yield-to-maturity is illustrated in Exhibit 6. As time passes, the Macaulay duration decreases smoothly but jumps upward by a small amount on coupon dates.

Exhibit 6:

Macaulay Duration between Coupon Payments with Constant Yield-to-Maturity for the BRWA 3.2% Bond



Question Set

1. For each pair of bonds, indicate the bond that *most likely* has greater interest rate risk:

First bond	Second bond	Which has greater interest rate risk?
4% coupon, paid semiannually, and five years	5% coupon, paid semiannually, and five years	

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First bond	Second bond	Which has greater interest rate risk?
to maturity, priced to yield 6%	to maturity, priced to yield 6%	
	4% coupon, paid semiannually, and six years to maturity, priced to yield 6%	
	5% coupon, paid semiannually, and five years to maturity, priced to yield 8%	
	4% coupon, paid semiannually, and five years to maturity, priced to yield 8%	

Hide Solution

Solution:

First bond	Second bond	Which has greater interest rate risk?
4% coupon, paid semiannually, and five years to maturity, priced to yield 6%	5% coupon, paid semiannually, and five years to maturity, priced to yield 6%	First bond
	4% coupon, paid semiannually, and six years to maturity, priced to yield 6%	Second bond
	5% coupon, paid semiannually, and five years to maturity, priced to yield 8%	First bond
	4% coupon, paid semiannually, and five years to maturity, priced to yield 8%	First bond

2. A bond analyst is comparing Bond D, a perpetual bond with a coupon of 5%, with Bond E, a zero-coupon bond maturing in five years. If both bonds are priced to yield 6%, Bond D has:

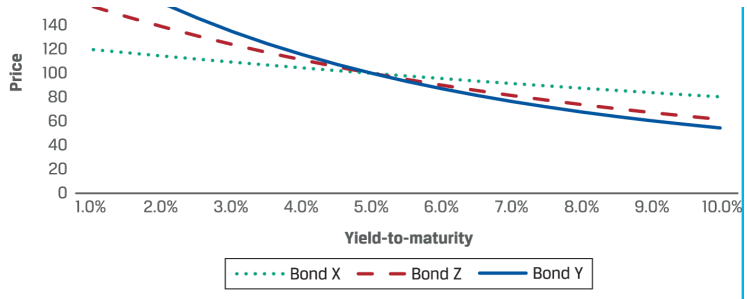
A. lower interest rate risk.
B. the same interest rate risk.
C. higher interest rate risk.

Hide Solution

Solution:

C is correct. The Macaulay duration for Bond D is 1.06/0.06 = 17.667. The Macaulay duration for Bond E is five years. The perpetual bond, Bond D, has higher interest rate risk.

3. The graph depicts the price–yield relationships for three bonds, Bond X, Bond Y, and Bond Z. The three bonds have the same coupon rate, 5%.



Which of the three bonds *most likely* has the longest time to maturity?

A. Bond X
B. Bond Y
C. Bond Z

Hide Solution

Solution:

B is correct. Bond Y has the steepest price-yield line, indicating that it has the greatest modified duration and, most likely, the longest time-to-maturity.

4. Between coupon payments, if the yield-to-maturity does not change, the Macaulay duration of a bond:

A. decreases throughout the coupon period.
B. is constant throughout the coupon period.
C. increases throughout the coupon period.

Hide Solution

Solution:

A is correct. During the coupon period, the Macaulay duration declines smoothly until the next coupon period, at which time it jumps. This is illustrated in Exhibit 6.

5. Two bonds have the same time-to-maturity and yield-to-maturity, but one is trading at a premium to par and the other at a discount. The Macaulay duration for the premium bond is *most likely*_____ that of the discount bond.

A. lower than
B. the same as
C. higher than

Hide Solution

Solution:

A is correct. If two bonds have the same maturity and the same yield-to-maturity but one trades at a premium and one trades at a discount to par, the premium bond must have a higher coupon rate. Therefore, the premium bond has the lower Macaulay duration.

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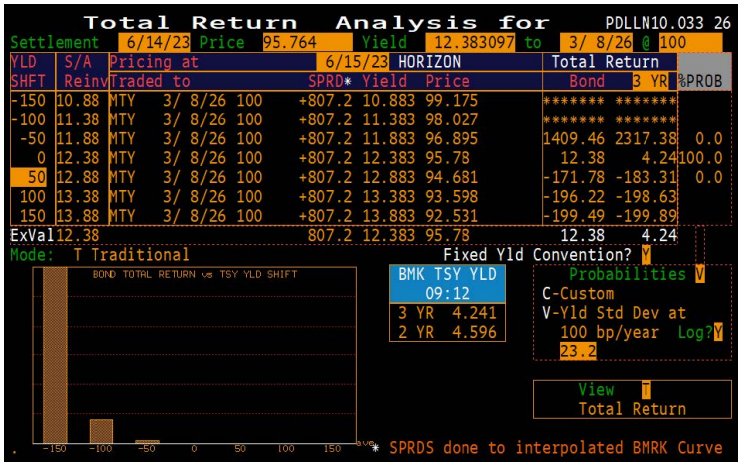

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regarding this conclusion, at some point, when the time-to-maturity is high enough, the Macaulay duration will exceed (1 + r)/r, reach a maximum, and then decrease. To me, for discount bonds, the MacDur will ALWAYS exceed (1+r)/r given that -(c-r) > 0, and thus the closed form expression = (1+r)/r + an always positive number (assuming c>0, r>0). Not sure what i got wrong

Created 4 months ago by Mingli Liang

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